DERIVED PULLBACK TO THE SMALL ETALE SITE IS AN EQUIVALENCE ON DERIVED CATEGORIES OF QUASI-COHERENT MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a module-level unit and counit check followed by an affine-cohomology computation of the derived pushforward and sheafification, closing both adjunction maps inside the delivered span

Editorial note. The label lemma-derived-quasi-coherent-small-etale-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The comparison of the small Zariski and etale sites and the sheaf property of the associated etale structure sheaf from Descent
- The equivalence of quasi-coherent modules on the small etale site with quasi-coherent modules on X, from Descent
- Vanishing of higher cohomology of quasi-coherent sheaves on affine objects, and the site-level lemma expressing cohomology over U through the top cohomology sheaf

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Prerequisite pointer invoked at proof line(s) 442; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 03DU: sources/stacks-chow/tags.txt, source lines 4433–4433. Prerequisite pointer invoked at proof line(s) 421; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 070S: sources/stacks-chow/tags.txt, source lines 8966–8966. Prerequisite pointer invoked at proof line(s) 415; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03DT: sources/stacks-chow/tags.txt, source lines 4432–4432. Prerequisite pointer invoked at proof line(s) 423; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 03DX: sources/stacks-chow/tags.txt, source lines 4436–4436. Prerequisite pointer invoked at proof line(s) 427; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Prerequisite pointer invoked at proof line(s) 439; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BKZ: sources/stacks-chow/tags.txt, source lines 14577–14577. Prerequisite pointer invoked at proof line(s) 444; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. The functor $\epsilon^* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_{étale})$ defined above is an equivalence.*

Proof. We will prove this by showing the functor $R\epsilon_* : D(\mathcal{O}_{étale}) \to D(\mathcal{O}_X)$ induces a quasi-inverse. We will use freely that ϵ_* is given by restriction to $X_{Zar} \subset X_{étale}$ and the description of $\epsilon^* = \mathrm{id}^*_{small,étale,Zar}$ in Descent, Lemma 070S.

For a quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ the adjunction map $\mathcal{F} \to \epsilon_*\epsilon^*\mathcal{F}$ is an isomorphism by the fact that $\mathcal{F}^a$ (Descent, Definition 03DU) is a sheaf as proved in Descent, Lemma 03DT. Conversely, every quasi-coherent $\mathcal{O}_{étale}$-module $\mathcal{H}$ is of the form $\epsilon^*\mathcal{F}$ for some quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$, see Descent, Proposition 03DX. Then $\mathcal{F} = \epsilon_*\mathcal{H}$ by what we just said and we conclude that the adjunction map $\epsilon^*\epsilon_*\mathcal{H} \to \mathcal{H}$ is an isomorphism for all quasi-coherent $\mathcal{O}_{étale}$-modules $\mathcal{H}$.

Let E be an object of $D_{QCoh}(\mathcal{O}_{étale})$ and denote $\mathcal{H}^q = H^q(E)$ its qth cohomology sheaf. Let $\mathcal{B}$ be the set of affine objects of $X_{étale}$. Then $H^p(U, \mathcal{H}^q) = 0$ for all $p > 0$, all $q \in \mathbf{Z}$, and all $U \in \mathcal{B}$, see Descent, Proposition 03DW and Cohomology of Schemes, Lemma 01XB. By Cohomology on Sites, Lemma 0BKZ this means that

$$H^q(U, E) = H^0(U, \mathcal{H}^q)$$

for all $U \in \mathcal{B}$. In particular, we find that this holds for affine opens $U \subset X$. It follows that the qth cohomology of $R\epsilon_*E$ over U is the value of the sheaf $\epsilon_*\mathcal{H}^q$ over U. Applying sheafification we obtain

$$H^q(R\epsilon_*E) = \epsilon_*\mathcal{H}^q$$

which in particular shows that $R\epsilon_*$ induces a functor $D_{QCoh}(\mathcal{O}_{étale}) \to D_{QCoh}(\mathcal{O}_X)$. Since ϵ^* is exact we then obtain $H^q(\epsilon^*R\epsilon_*E) = \epsilon^*\epsilon_*\mathcal{H}^q = \mathcal{H}^q$ (by discussion above). Thus the adjunction map $\epsilon^*R\epsilon_*E \to E$ is an isomorphism.

Conversely, for $F \in D_{QCoh}(\mathcal{O}_X)$ the adjunction map $F \to R\epsilon_*\epsilon^*F$ is an isomorphism for the same reason, i.e., because the cohomology sheaves of $R\epsilon_*\epsilon^*F$ are isomorphic to $\epsilon_*H^m(\epsilon^*F) = \epsilon_*\epsilon^*H^m(F) = H^m(F)$. □